

```
In [28]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

```
In [29]: df = pd.read_csv(r"D:\python\EDA Code Alpha\Sample Superstore.csv", encoding='latin1')
df.head()
```

```
Out[29]:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States

5 rows × 21 columns



```
In [30]: df.shape
```

```
Out[30]: (9994, 21)
```

```
In [31]: df.columns
```

```
Out[31]: Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',  
              'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',  
              'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',  
              'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],  
             dtype='str')
```

```
In [32]: df.dtypes
```

```
Out[32]: Row ID          int64  
Order ID          str  
Order Date        str  
Ship Date         str  
Ship Mode         str  
Customer ID       str  
Customer Name     str  
Segment           str  
Country           str  
City              str  
State             str  
Postal Code       int64  
Region            str  
Product ID        str  
Category          str  
Sub-Category      str  
Product Name      str  
Sales             float64  
Quantity          int64  
Discount          float64  
Profit            float64  
dtype: object
```

```
In [33]: df.info()  
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                 9994 non-null   int64
1   Order ID               9994 non-null   str
2   Order Date             9994 non-null   str
3   Ship Date              9994 non-null   str
4   Ship Mode               9994 non-null   str
5   Customer ID            9994 non-null   str
6   Customer Name          9994 non-null   str
7   Segment                9994 non-null   str
8   Country                 9994 non-null   str
9   City                   9994 non-null   str
10  State                  9994 non-null   str
11  Postal Code            9994 non-null   int64
12  Region                 9994 non-null   str
13  Product ID             9994 non-null   str
14  Category                9994 non-null   str
15  Sub-Category           9994 non-null   str
16  Product Name           9994 non-null   str
17  Sales                   9994 non-null   float64
18  Quantity                9994 non-null   int64
19  Discount                9994 non-null   float64
20  Profit                  9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB
```

Out[33]:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

In [34]:

```
df.isnull().sum()
```

```
Out[34]: Row ID          0
         Order ID       0
         Order Date     0
         Ship Date      0
         Ship Mode       0
         Customer ID    0
         Customer Name   0
         Segment        0
         Country        0
         City           0
         State          0
         Postal Code    0
         Region         0
         Product ID     0
         Category       0
         Sub-Category   0
         Product Name   0
         Sales          0
         Quantity       0
         Discount       0
         Profit         0
         dtype: int64
```

```
In [35]: df.duplicated().sum()
```

```
Out[35]: np.int64(0)
```

```
In [36]: df['Order Date'] = pd.to_datetime(df['Order Date'])
         df['Ship Date'] = pd.to_datetime(df['Ship Date'])
```

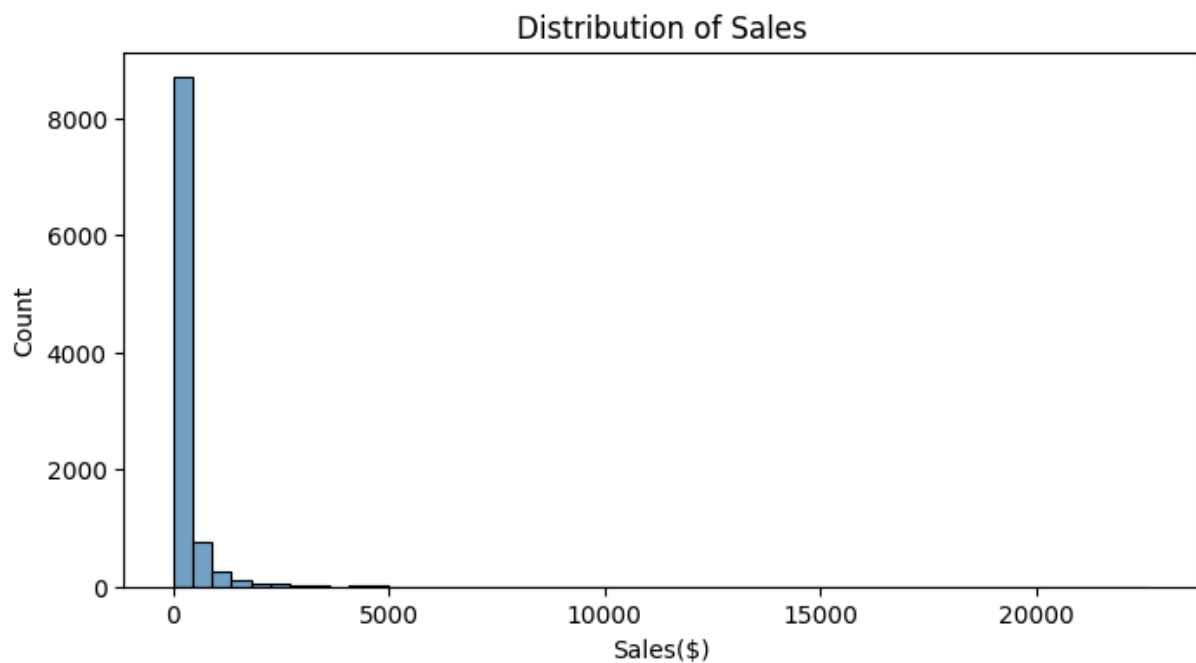
```
In [37]: df.dtypes
```

```
Out[37]: Row ID          int64
         Order ID       str
         Order Date     datetime64[us]
         Ship Date      datetime64[us]
         Ship Mode       str
         Customer ID    str
         Customer Name   str
         Segment        str
         Country        str
         City           str
         State          str
         Postal Code    int64
         Region         str
         Product ID     str
         Category       str
         Sub-Category   str
         Product Name   str
         Sales          float64
         Quantity       int64
         Discount       float64
         Profit         float64
         dtype: object
```

```
In [38]: df['Sales'].describe()
```

```
Out[38]: count    9994.000000
mean       229.858001
std        623.245101
min         0.444000
25%        17.280000
50%        54.490000
75%       209.940000
max      22638.480000
Name: Sales, dtype: float64
```

```
In [41]: plt.figure(figsize=(8,4))
sns.histplot(df['Sales'], bins=50, color='steelblue')
plt.title('Distribution of Sales')
plt.xlabel('Sales($')
plt.ylabel('Count')
plt.show()
```

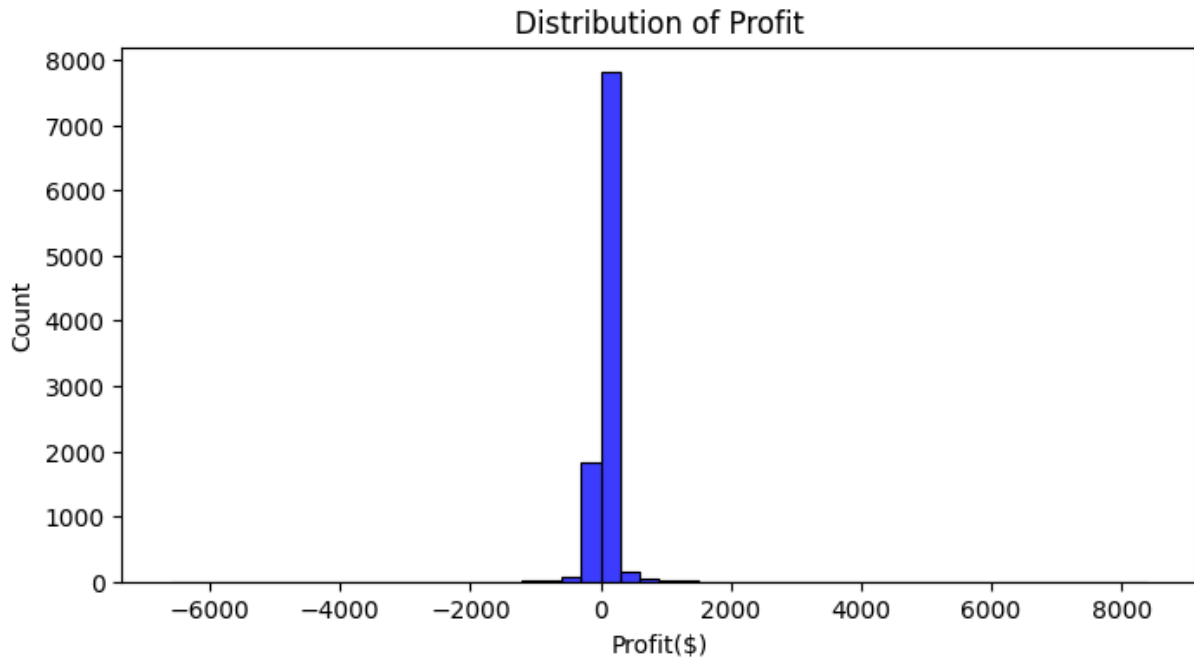


```
In [42]: df['Profit'].describe()
```

```
Out[42]: count    9994.000000
mean       28.656896
std        234.260108
min      -6599.978000
25%         1.728750
50%         8.666500
75%        29.364000
max       8399.976000
Name: Profit, dtype: float64
```

```
In [44]: plt.figure(figsize=(8,4))
sns.histplot(df['Profit'], bins=50, color='blue')
plt.title('Distribution of Profit')
plt.xlabel('Profit($')
plt.show()
```

```
plt.ylabel('Count')
plt.show()
```



```
In [45]: df['Cost'] = df['Sales'] - df['Profit']
```

```
In [47]: df[['Product Name', 'Sales', 'Profit', 'Cost']].head(10)
df[['Product Name', 'Sales', 'Profit', 'Cost']].tail(10)
```

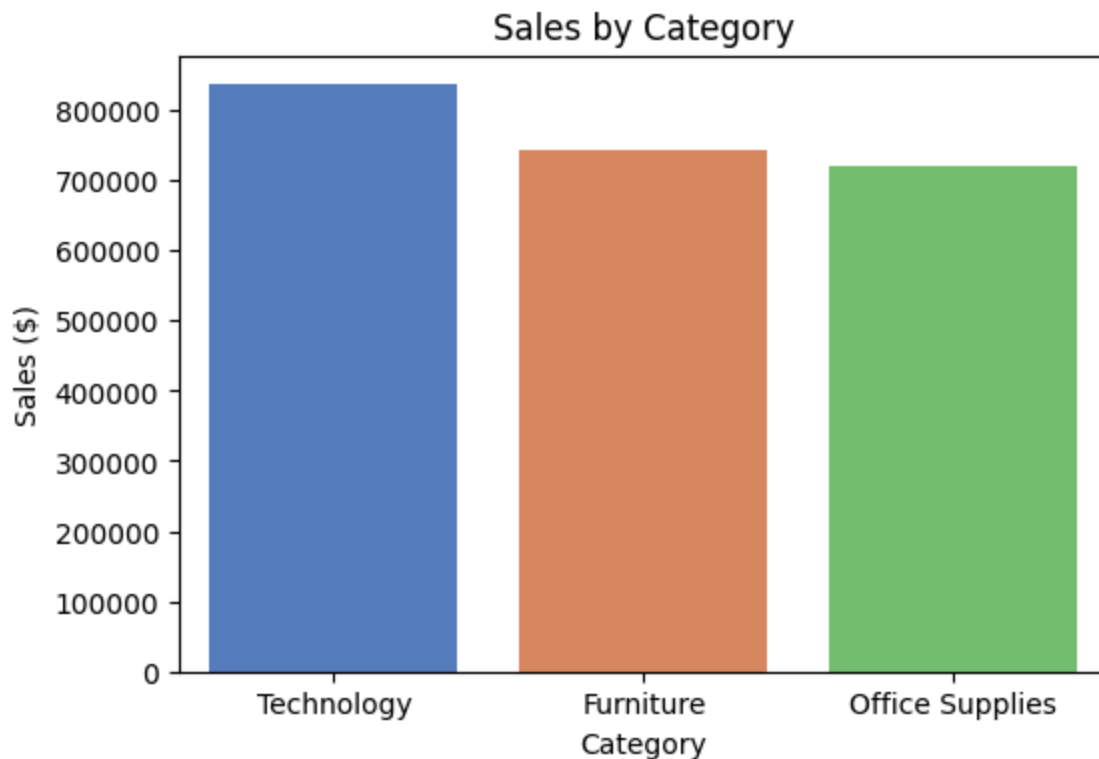
```
Out[47]:
```

	Product Name	Sales	Profit	Cost
9984	Self-Adhesive Removable Labels	31.500	15.1200	16.3800
9985	Acme Hot Forged Carbon Steel Scissors with Nic...	55.600	16.1240	39.4760
9986	Memorex Mini Travel Drive 64 GB USB 2.0 Flash ...	36.240	15.2208	21.0192
9987	Logitech G430 Surround Sound Gaming Headset wi...	79.990	28.7964	51.1936
9988	Panasonic KX - TS880B Telephone	206.100	55.6470	150.4530
9989	Ultra Door Pull Handle	25.248	4.1028	21.1452
9990	Tenex B1-RE Series Chair Mats for Low Pile Car...	91.960	15.6332	76.3268
9991	Aastra 57i VoIP phone	258.576	19.3932	239.1828
9992	It's Hot Message Books with Stickers, 2 3/4" x 5"	29.600	13.3200	16.2800
9993	Acco 7-Outlet Masterpiece Power Center, Wihtou...	243.160	72.9480	170.2120

```
In [48]: category_sales = df.groupby('Category')['Sales'].sum().sort_values(ascending=False)
print(category_sales)
```

```
Category
Technology      836154.0330
Furniture        741999.7953
Office Supplies  719047.0320
Name: Sales, dtype: float64
```

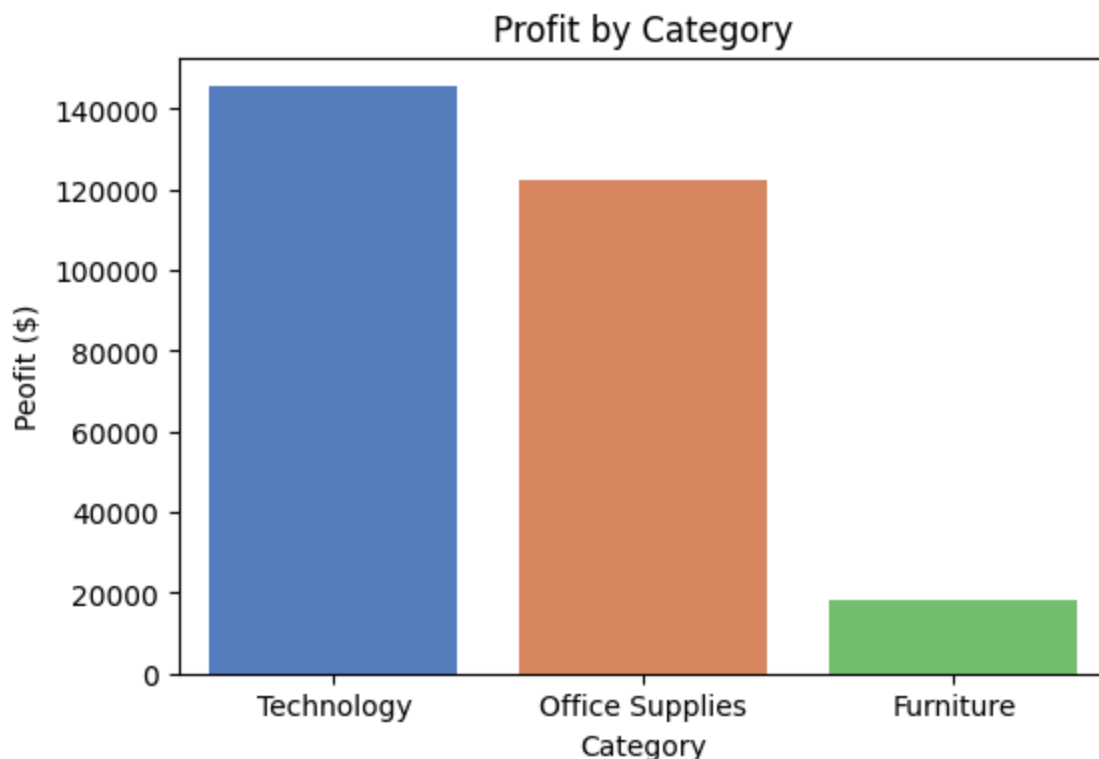
```
In [49]: plt.figure(figsize=(6, 4))
sns.barplot(x=category_sales.index, y=category_sales.values, palette='muted')
plt.title('Sales by Category')
plt.xlabel('Category')
plt.ylabel('Sales ($)')
plt.show()
```



```
In [50]: category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascending=False)
print(category_profit)
```

```
Category
Technology      145454.9481
Office Supplies  122490.8008
Furniture        18451.2728
Name: Profit, dtype: float64
```

```
In [51]: plt.figure(figsize=(6, 4))
sns.barplot(x=category_profit.index, y=category_profit.values, palette='muted')
plt.title('Profit by Category')
plt.xlabel('Category')
plt.ylabel('Profit ($)')
plt.show()
```



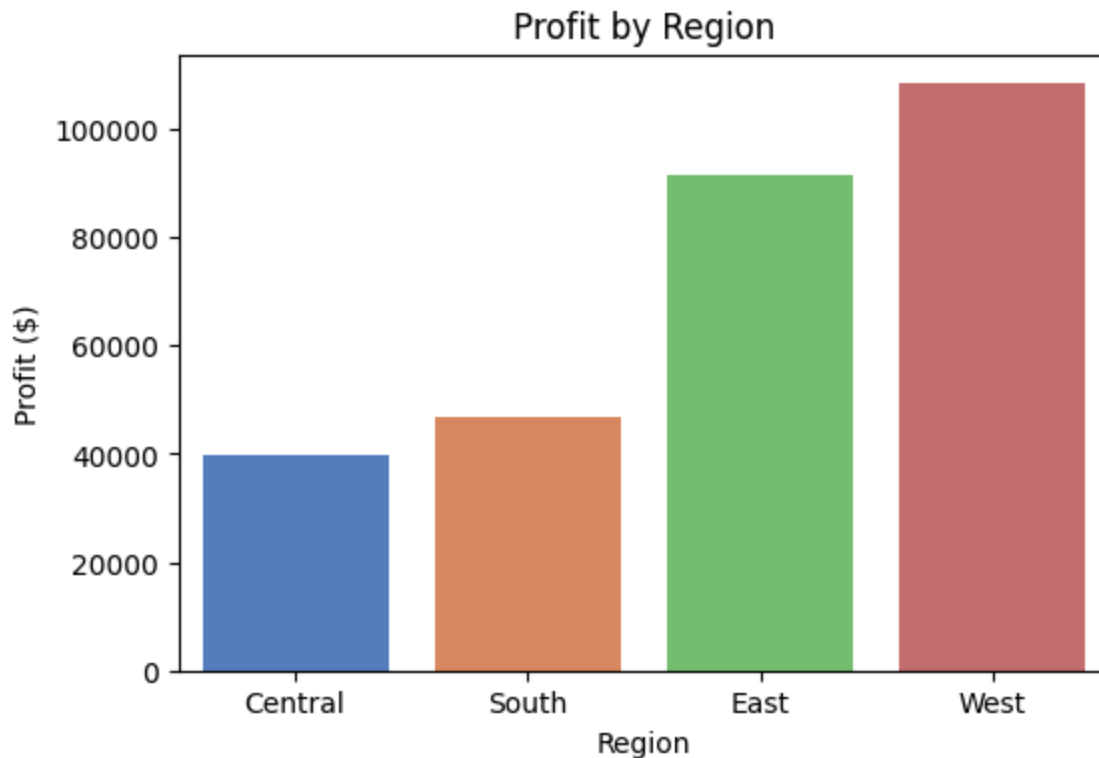
```
In [54]: category_discount = df.groupby('Category')['Discount'].mean().sort_values()
print(category_discount)
```

```
Category
Technology      0.132323
Office Supplies  0.157285
Furniture       0.173923
Name: Discount, dtype: float64
```

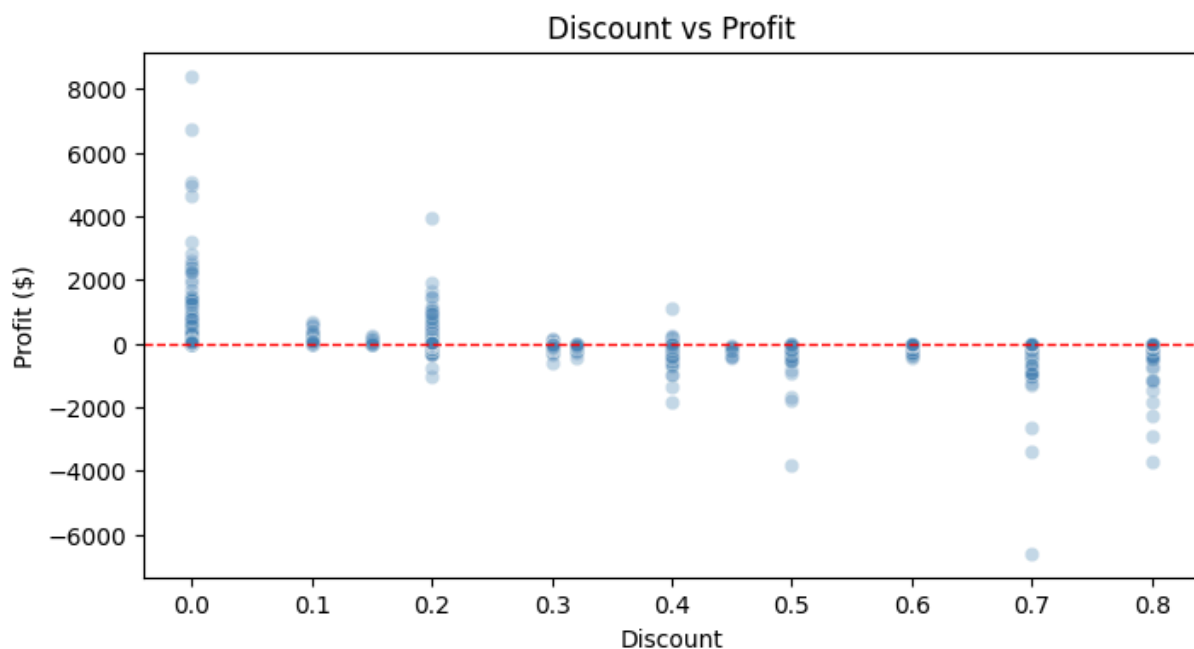
```
In [55]: region_profit = df.groupby('Region')['Profit'].sum().sort_values()
print(region_profit)
```

```
Region
Central    39706.3625
South      46749.4303
East       91522.7800
West       108418.4489
Name: Profit, dtype: float64
```

```
In [56]: plt.figure(figsize=(6, 4))
sns.barplot(x=region_profit.index, y=region_profit.values, palette='muted')
plt.title('Profit by Region')
plt.xlabel('Region')
plt.ylabel('Profit ($)')
plt.show()
```



```
In [57]: plt.figure(figsize=(8, 4))
sns.scatterplot(x=df['Discount'], y=df['Profit'], alpha=0.3, color='steelblue')
plt.title('Discount vs Profit')
plt.xlabel('Discount')
plt.ylabel('Profit ($)')
plt.axhline(0, color='red', linewidth=1, linestyle='--')
plt.show()
```



Key Insights — Superstore EDA

1. Sales

- Technology is the top-selling category with \$836K
- Most orders have a small value — below \$54

2. Profitability

- Technology is the most profitable category with \$145K
- Furniture has *742K in sales but only* 18K in profit

3. Discounts

- Furniture has the highest average discount at 17.3% which explains its low profit
- Discounts above 40% turn orders into losses

4. Regions

- West is the most profitable region with \$108K
- Central is the weakest region with only \$39K